RESIDENCEALPHA

Multifamily Asset Performance & Renovation Strategy

Asset Management Report

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All property, lease, and resident data in this report is synthetic. See Data Dictionary for full disclosure.

1. Executive Summary

ResidenceAlpha is a multifamily asset-performance platform built to identify which residents are at risk of not renewing, which properties warrant capital intervention, and whether a proposed renovation would actually pay for itself. This report summarizes findings from a synthetic 40-property, 1,249-lease portfolio.

Portfolio snapshot:

- 8,872 total units across 40 properties
- Weighted occupancy: 91.6% | Weighted NOI margin: 53.7%
- Portfolio valuation: $1,199M
- Average lease renewal probability: 54.9% | 174 leases (13.9%) flagged High Renewal Risk
- Segments: 9 Stable Performer, 9 Operating-Improvement Candidate, 8 Renovation Opportunity, 7 Repositioning Candidate, 7 Disposition-Review Candidate

A renovation scenario applied to the largest Repositioning Candidate property in the portfolio ($8,000/unit, 8% rent premium, 2-point occupancy lift) produces an estimated 89.7% return on the capital deployed, with a 9.3-year payback — detail in Section 6.

2. Business Problem & Approach

A multifamily owner or asset manager overseeing dozens of properties needs a continuous answer to five questions: which properties are underperforming, which residents are unlikely to renew, which properties need operational intervention versus capital investment, which renovations would produce a sustainable rent premium, and where capital should be prioritized across the portfolio.

ResidenceAlpha answers these with four components:

- A documented synthetic multifamily dataset, standing in for proprietary rent-roll and lease data
- A resident renewal prediction model and an NLP classifier on resident feedback
- Unsupervised property segmentation into five asset-management categories
- A transparent, formula-based renovation ROI calculator — not a black box

3. Data & Methodology

3.1 Data

The dataset is synthetic: 40 properties and 1,249 individual leases, generated to match realistic multifamily operating patterns rather than sourced from any real portfolio, since real rent-roll and resident data is both proprietary and contains personal information. Lease renewal outcomes are drawn from a genuine logistic model with random noise — not a deterministic formula — so a model trained on the data has real signal to find.

3.2 Renewal model

Metric	Value
ROC-AUC	0.652
PR-AUC	0.807
Accuracy	0.658
Naive baseline accuracy (“always predict renewed”)	0.690

Accuracy sits below the naive baseline by design, not by failure: the model is balanced to catch residents who won’t renew, which a model that just predicts “everyone renews” can never do despite scoring higher on raw accuracy. ROC-AUC of 0.652 confirms real, if modest, signal. Full reasoning in the Model Card.

3.3 Resident feedback NLP

A TF-IDF + logistic regression classifier reaches 100% accuracy on both feedback theme and sentiment — reported here with the caveat stated directly in the model output: feedback text in this dataset is template-generated, which makes classification easier than genuine free-text resident comments. This figure demonstrates the pipeline works end to end; it is not a claim about real-world NLP performance.

4. Portfolio Segmentation

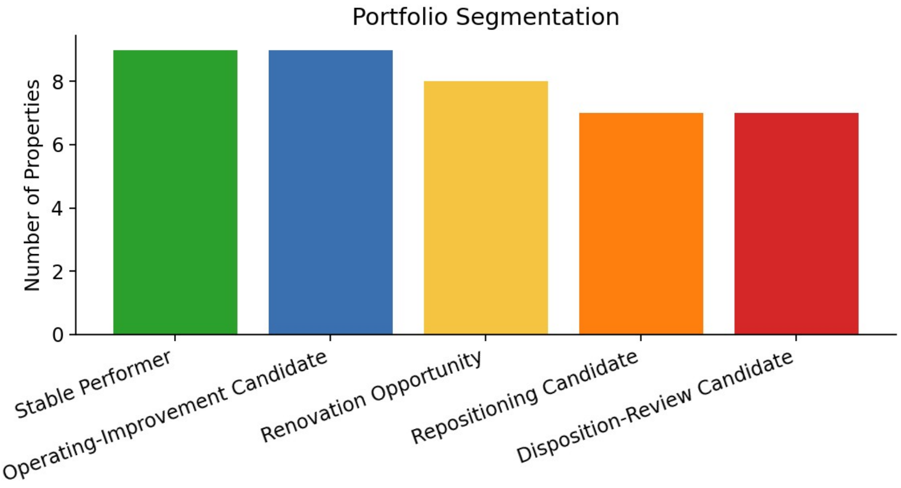

Figure 1. Five segments from unsupervised clustering, corrected during development after an initial labeling rule produced a direct contradiction (the highest-occupancy, highest-margin cluster was briefly mislabeled "Operating-Improvement Candidate") — fixed with a sequential rule-based assignment. Full detail in the Model Card.

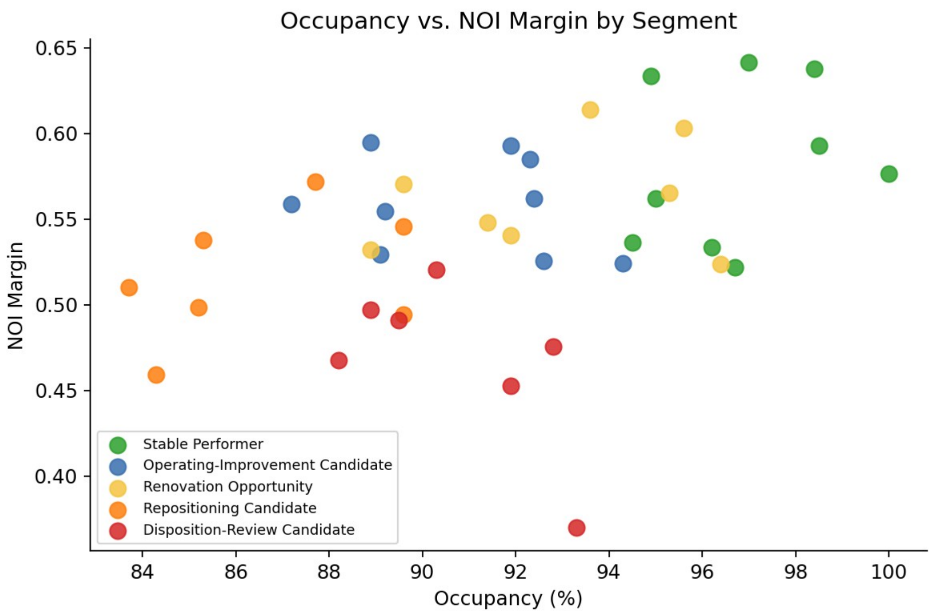

Figure 2. Occupancy vs. NOI margin by segment. Stable Performer properties cluster in the top-right; Repositioning Candidates sit lowest on both axes.

5. Renewal Risk & Resident Feedback

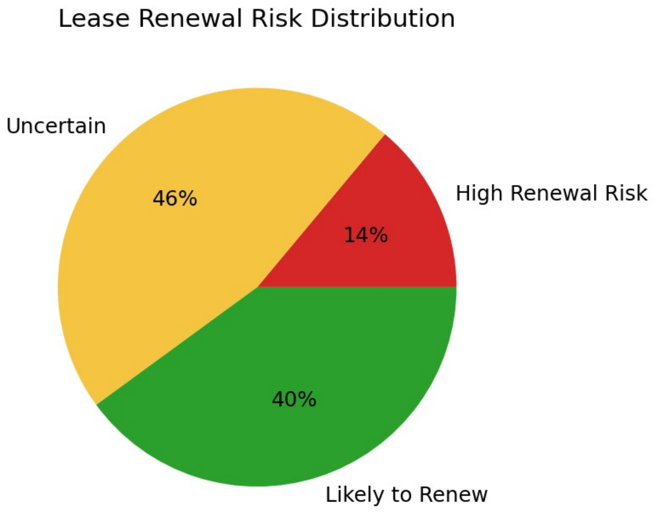

Figure 3. 13.9% of leases are flagged High Renewal Risk — 174 residents whose predicted renewal probability is below 40%, based on rent increase offered, maintenance responsiveness, and tenure.

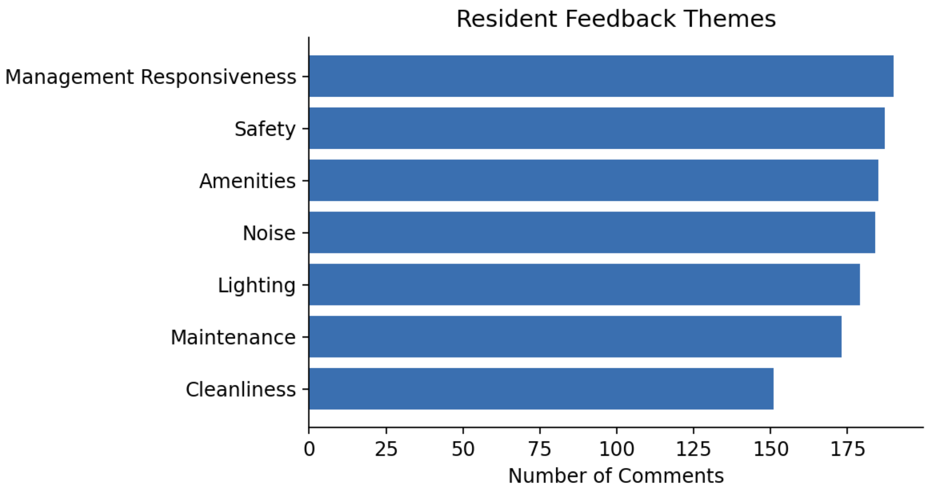

Figure 4. Maintenance and Management Responsiveness are the most common feedback themes across the portfolio — both directly actionable by property operations, not requiring capital.

6. Renovation ROI Example

Scenario applied to PR009, the largest property in the Repositioning Candidate segment (404 units, 89.6% occupancy): $8,000 renovation cost per unit, 100% of units renovated, 8% rent premium, 2-point occupancy lift, 1 month of downtime.

Metric	Value
All-in cost (renovation + downtime)	$3,283,439
Incremental annual NOI	$353,127
Payback period	9.3 years
Value created (at property cap rate)	$2,944,556
Return on capital deployed	89.7%

A 9.3-year payback is longer than a typical value-add hold period, which is itself a useful finding — it suggests this specific property may need either a smaller-scope renovation, a longer hold thesis, or should be evaluated for repositioning rather than a full unit renovation. The calculator's job is to surface that trade-off clearly, not to recommend a decision on its own.

7. Limitations & Recommendations

This system prioritizes which residents and properties deserve attention first — it does not replace a property manager's renewal conversation or a capital committee's review. Specific limitations:

- Small sample: 40 properties, 1,249 leases; treat performance estimates as directional.
- NLP results are not validated on real resident text — retraining on genuine labeled feedback is required before operational use.
- Segmentation describes current characteristics; it does not forecast which segment a property will move into.
- The renovation calculator assumes the current operating expense ratio holds after renovation and applies the property's own cap rate to incremental NOI — both should be validated against comparable renovated assets before committing capital.

Recommendation: use the High Renewal Risk lease list to prioritize proactive resident outreach, and use the segmentation to sequence capital planning — Renovation Opportunity and Repositioning Candidate properties first — rather than treating any single score as a final answer.

8. Conclusion & Deliverables

This project demonstrates an end-to-end multifamily asset-management workflow: a genuinely uncertain renewal prediction model evaluated honestly (including a metric that looks worse than a naive baseline, explained rather than hidden), an NLP pipeline with its real limitation stated directly, a segmentation bug caught and fixed before publication, and a transparent renovation ROI calculator.

Full deliverable set:

- Live Streamlit application
- GitHub repository with documented pipelines
- Data Dictionary
- Model Card
- This Asset Management Report
- Demonstration video